

GAUGE-ALIGNED ENDPOINT INTERFACES FOR CROSS-DIMENSIONAL TEXT EMBEDDING DISTILLATION

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ABSTRACT

Knowledge distillation can compress text embedding models, but directly matching teacher and student vectors is difficult when their dimensions and coordinate systems differ. We propose GATE-KD, a final-embedding method that first reduces teacher embeddings to the student dimension with a fixed PCA projection and then fits their orientation to the student using closed-form orthogonal Procrustes alignment. This produces pointwise training targets from cached teacher embeddings without a trainable alignment module or intermediate teacher states. As an auxiliary signal, GATE-KD also matches degree-zero persistence summaries, namely distributions of connectivity scales, in the original teacher and student spaces; this signal remains well-defined across different dimensions. Under a label-free but task-adapted protocol spanning three teacher–student configurations and nine sentence-level benchmarks, GATE-KD achieves the highest numerical average among the evaluated distillation methods, improving over the strongest baseline by 0.38–1.59 points. Controlled experiments show that a per-minibatch rotation-invariant baseline trails GATE-KD by 2.05 points; separate schedule controls associate most of this gap with a corpus-level fitted gauge rather than repeated refitting, while H_0 adds 0.53 points over endpoint-only training.

1 INTRODUCTION

Text embedding models support retrieval, semantic similarity, clustering, and classification, but the strongest encoders are often too costly for latency- or memory-constrained deployment. Knowledge distillation offers a natural route to compact embedding models by transferring supervision from a frozen teacher to a smaller student (Hinton et al., 2015; Reimers & Gurevych, 2020). Unlike class logits, however, embedding coordinates have no shared semantics across models. Teacher and student representations may differ in dimension, orientation, and architecture. Existing methods address this mismatch with learned projectors, relational or contrastive objectives, and intermediate-layer supervision (Park et al., 2019; Tian et al., 2020; Xu et al., 2023; Zhang et al., 2024a; Truong et al., 2025; Dao et al., 2026). Hierarchy-based transfer can be effective, but it requires access to internal teacher representations and assumptions about cross-model layer correspondence (Ko et al., 2023; Yu et al., 2025). This machinery can obscure a more basic question: whether teacher and student endpoints have first been expressed in comparable coordinates. We therefore study the endpoint interface as a potential bottleneck in cross-dimensional distillation.

We test this hypothesis by centering the transfer mechanism on an *extrinsic interface* that expresses teacher representations in coordinates the student can match pointwise. We augment this primary mechanism with an *intrinsic regularizer* defined on properties of the representation cloud that do not depend on its coordinates or ambient dimension. This decomposition leads to **GATE-KD**, Gauge Alignment and Topology for Embedding Distillation. GATE-KD fits a fixed principal-component subspace to cached teacher endpoints to provide a non-trainable reduction at the student width. Within the retained subspace, orthogonal Procrustes alignment fits the orientation of the teacher targets to the student’s initial coordinates in closed form, without changing sample correspondence or pairwise geometry. Because projection cannot preserve all teacher geometry, GATE-KD adds a degree-zero persistent-homology (H_0) summary regularizer that matches distributions of connectivity scales in the native teacher and student spaces. This coarse auxiliary summary is invariant

to coordinates and ambient dimension; it does not preserve sample-level neighborhoods or the full metric geometry. The resulting objective supervises only the final student embedding, trains no alignment module, and requires no online teacher forward pass during ordinary training.

Experiments across three teacher–student configurations and nine sentence-level benchmarks under a label-free task-adapted protocol provide evidence consistent with this endpoint-centered view. A per-minibatch rotation-invariant control trails GATE-KD by 2.05 points; separate schedule controls suggest that consistent corpus-level targets matter beyond allowing an adaptive rotation, while periodic refitting provides only a smaller additional gain. Native-space H_0 adds 0.53 points over endpoint-only training and performs best among the evaluated structural controls and weights. Overall, GATE-KD obtains the numerically highest aggregate score among the evaluated distilled students in all three configurations, exceeding the next-best baseline by 0.38, 0.48, and 1.59 points. Within the measured minibatch optimization loop, it also records the lowest update time and cumulative optimization time among the evaluated methods in all three configurations. These results show that final-embedding-only distillation can be competitive with the evaluated hierarchy-based alternatives in the task-adapted sentence-level settings studied here.

We make three contributions.

- We formulate cross-dimensional embedding distillation around a comparable final-embedding interface, separating pointwise coordinate comparability from auxiliary coordinate-invariant structure.
- We introduce a fixed-reduction, closed-form gauge interface using only cached teacher endpoints, together with native-space H_0 regularization as a complementary signal and epoch-wise gauge refresh as a secondary optimization refinement.
- We provide a matched evaluation and controlled analyses that isolate gauge fitting, subspace construction, refitting, and H_0 connectivity-scale matching, clarifying which components account for the observed gains and the regime in which the conclusions apply.

2 RELATED WORK

Early text-embedding distillation regressed student sentence vectors directly onto teacher outputs, demonstrating dense supervision from unlabeled text (Reimers & Gurevych, 2020); later methods combine endpoint transfer with contrastive, relative-geometric, multilingual, or specialized objectives (Gao et al., 2021; Xu et al., 2023; Xie et al., 2023; Mao & Nakagawa, 2023; Kim et al., 2023; Zhang et al., 2024a; Li et al., 2025; Vujanic & Rückstieß, 2026) and with generated or task-targeted data (Lee et al., 2024; Akram et al., 2026). A separate family exploits the representation hierarchy: EMO aligns token-level relations and hidden states, TALAS combines a teacher endpoint anchor with lower-layer self-distillation, and others transfer mixtures of layers, intermediate states, or attention (Truong et al., 2025; Dao et al., 2026; Vu et al., 2026; Wang et al., 2024). Such supervision relies on access to internal representations and on cross-model layer correspondence, whose robustness has been questioned (Ko et al., 2023; Yu et al., 2025). GATE-KD studies the complementary terminal-only regime using cached final teacher embeddings. Because evaluation settings differ across these systems, with EMO using task-specific supervised training and TALAS a label-free task-adapted corpus, we follow the TALAS data and evaluation protocol for a like-for-like comparison; this controls corpus and task coverage but does not turn source-task results into corpus-disjoint transfer.

Cross-dimensional endpoint supervision additionally requires a common output space. Learned projection layers are widely used, but their parameterization and normalization can materially alter student optimization (Zhao et al., 2022; Miles & Mikolajczyk, 2024; Chen et al., 2023). TCS instead constructs reduced coordinates from cached teacher features using PCA (Zhou et al., 2025); the unaligned row of Table 3 plays the role of this fixed-PCA target in our matched comparison. After reduction, orthogonal changes of basis preserve the retained inner products, so representation-similarity work treats them as natural equivalences (Kornblith et al., 2019; Williams et al., 2021; Maystre et al., 2025; Han, 2026), and Bhattarai et al. (2025) show that unconstrained projected matching can reach zero feature loss without preserving teacher Gram geometry, a shortcut Procrustes avoids. Dimension consistency is also studied within a single model: Matryoshka representation learning trains nested prefixes of one embedding to remain useful at several truncation ranks

(Kusupati et al., 2022), and MIPIC adds self-distillation between full and truncated embeddings with intra-relational objectives (Huy, 2026). There the full embedding is its own teacher; GATE-KD instead transfers a frozen external teacher of a different width, where no shared parameterization links the two spaces and a coordinate interface must be constructed. It combines a fixed PCA bottleneck with a fitted, closed-form Procrustes gauge rather than a trainable projector; the target-rank sensitivity in Appendix C.3 can be read as treating the student width as a nested target rank of the teacher subspace.

Coordinate-free objectives offer a complementary route to transferring structure. Relational, similarity-preserving, and contrastive distillation match pairwise distances, Gram matrices, or positive-negative relations (Park et al., 2019; Tung & Mori, 2019; Tian et al., 2020); related methods impose graph- or geometry-aware constraints in graph, dialogue, and diffusion settings (Joshi et al., 2024; Burykina et al., 2025; Zhou et al., 2026). Persistent homology extends this view to multiscale point-cloud structure: TopKD transfers topology through approximate persistence images, topological-guidance distillation uses persistence features from multiple teachers, and topologically regularized embeddings impose prior topology on learned low-dimensional representations (Kim et al., 2024; Jeon et al., 2024; Heiter et al., 2023). GATE-KD uses topology only as a dimension- and coordinate-invariant auxiliary, matching sorted degree-zero persistence death times in the native teacher and student spaces, while the explicit endpoint interface supplies the primary correspondence signal.

3 METHOD

Let $f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}$ be a frozen teacher and $f_S(\cdot; \theta) : \mathcal{X} \rightarrow \mathbb{R}^{d_S}$ a student with parameters θ and $d_T > d_S$. We represent embeddings as row vectors throughout. We run the teacher once over an unlabeled corpus $\{x_i\}_{i=1}^N$ and cache its normalized final sentence embeddings

$$\mathbf{t}_i = \text{norm}(f_T(x_i)) \in \mathbb{S}^{d_T-1}, \quad (1)$$

where $\text{norm}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$. The corresponding student embedding is $\mathbf{z}_i(\theta) = \text{norm}(f_S(x_i; \theta)) \in \mathbb{S}^{d_S-1}$; we suppress θ when it is clear from context. GATE-KD primarily transfers teacher knowledge through an extrinsic interface that expresses each teacher endpoint in coordinates the student can match pointwise. It augments this interface with an intrinsic regularizer that requires neither shared coordinates nor equal ambient dimension.

3.1 EXTRINSIC ENDPOINT INTERFACE

Subspace selection. Let $\mu = N^{-1} \sum_i \mathbf{t}_i$. We fit the leading d_S principal directions of the centered cache and collect them as $P_{\text{PCA}} \in \mathbb{R}^{d_T \times d_S}$, with $P_{\text{PCA}}^\top P_{\text{PCA}} = I_{d_S}$. Centering is used to estimate the principal directions; in the evaluated recipe, the cached vectors are not centered when the projection is applied. The normalized coordinates before gauge alignment are

$$\mathbf{y}_i = \text{norm}(\mathbf{t}_i P_{\text{PCA}}) \in \mathbb{S}^{d_S-1}. \quad (2)$$

The PCA subspace is fitted once on the full teacher cache and remains fixed throughout training. Centering the cache to estimate covariance directions and subtracting the mean when applying the interface are distinct operations: the former selects a subspace, whereas the latter would make the target transform affine and alter its angular geometry after row normalization. We use only the former, preserving a linear endpoint interface. Appendix C.2 evaluates both choices together with whitening and random-subspace controls.

Gauge alignment. The basis used to express the retained subspace affects a pointwise loss even though it does not affect the geometry within that subspace. We resolve this coordinate ambiguity with orthogonal Procrustes alignment (Schönemann, 1966; Bhattarai et al., 2025). We treat the orientation as a nuisance coordinate variable rather than as teacher content. A gauge-fit set $\mathcal{C}$ is selected before training and reused unchanged; in the main experiments it is the complete distillation corpus. Let $Y_{\mathcal{C}}$ contain $\mathbf{y}_i$ as rows and let $Z_{\mathcal{C}}^{(e)}$ contain the corresponding student embeddings $\mathbf{z}_i^{(e)}$

at the start of epoch e . We compute

$$\begin{aligned} U\Sigma V^\top &= \text{svd}\left(Y_C^\top Z_C^{(e)}\right), \\ R^{(e)} &= UV^\top \in \arg \min_{R \in O(d_S)} \left\| Y_C R - Z_C^{(e)} \right\|_F^2. \end{aligned} \quad (3)$$

The initialized student determines $R^{(0)}$. After each non-final epoch, we re-encode the same gauge-fit sentences and refit the gauge for the following epoch. The target used within epoch e is therefore

$$\tau_i^{(e)} = y_i R^{(e)}. \quad (4)$$

Each target is fixed within an epoch. The refit is a closed-form coordinate update rather than a learned projector. The minimizer in Equation 3 is unique exactly when the cross-covariance $Y_C^\top Z_C^{(e)}$ has full rank (Schönemann, 1966); in directions with vanishing singular values, $UV^\top$ is one of several minimizers and the returned orientation depends on the SVD convention.

Relation to the endpoint objective. Because the rows of Y_C and $Z_C^{(e)}$ have unit norm, their cosine endpoint objective satisfies

$$\frac{1}{|C|} \sum_{i \in C} \left(1 - \langle \mathbf{z}_i^{(e)}, \mathbf{y}_i R \rangle\right) = \frac{1}{2|C|} \left\| Z_C^{(e)} - Y_C R \right\|_F^2. \quad (5)$$

Proposition 1 (Optimality and invariance of the fitted gauge). *For row-normalized Y_C and $Z_C^{(e)}$, the update $R^{(e)}$ in Equation 3 globally minimizes the gauge-fit cosine endpoint loss over $O(d_S)$. Moreover, for every projected target matrix Y and every $R \in O(d_S)$,*

$$(YR)(YR)^\top = YY^\top. \quad (6)$$

Thus fitting the gauge changes the coordinate presentation but not sample correspondence, pairwise cosines, chord distances, or persistent-homology signatures of the projected teacher targets.

The proof is given in Appendix A.

PCA supplies a fixed student-width subspace, while Procrustes determines how that fixed geometry is presented in the student’s coordinates. The update does not select a new subspace or adapt the teacher’s within-target geometry. At each refit boundary, with the student fixed, it cannot increase the gauge-fit endpoint loss and leaves the native-space H_0 term unchanged. This conditional guarantee does not imply monotonic epoch-to-epoch training because the student changes between refits.

3.2 NATIVE-SPACE H_0 PERSISTENCE-SUMMARY REGULARIZATION

The endpoint interface can reproduce only the geometry retained after projection. Although P_{PCA} has orthonormal columns, $P_{\text{PCA}} P_{\text{PCA}}^\top$ is a rank- d_S projector, so the cosine Gram matrix of the normalized coordinates in Equation 2 need not equal that of the original teacher embeddings. This loss of geometry is unavoidable whenever the normalized teacher cache has rank above the student width.

Proposition 2 (Rank-relaxed lower bound on Gram distortion). *Let $T \in \mathbb{R}^{N \times d_T}$ collect the normalized teacher cache embeddings, with singular values $\sigma_1(T) \geq \dots \geq \sigma_r(T) > 0$. For every $Z \in \mathbb{R}^{N \times d_S}$ collecting student embeddings of the same N sentences,*

$$\left\| TT^\top - ZZ^\top \right\|_F^2 \geq \sum_{j > d_S} \sigma_j(T)^4. \quad (7)$$

Thus exact preservation of the teacher cosine Gram matrix is impossible when $\text{rank}(T) > d_S$. The bound relaxes the unit-diagonal constraint on $ZZ^\top$ and therefore need not be tight for normalized student embeddings.

The proof is given in Appendix A.

Rather than attempting to reconstruct the full teacher Gram geometry, which is generally impossible at student rank, we match a lower-complexity, coordinate-free summary of its connectivity scales. We therefore add a persistence-summary regularizer defined directly in the teacher’s native space.

For a minibatch $\mathcal{B} \subseteq \{1, \dots, N\}$ of size $|\mathcal{B}| = B$, we use the chord distance

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u}, \mathbf{v} \rangle}. \quad (8)$$

We parameterize the Vietoris–Rips filtration by the distance threshold: an edge enters at scale a when its distance is at most a . The $B - 1$ finite death times of degree-zero Vietoris–Rips persistence are the edge weights of the point cloud’s minimum spanning tree (Hofer et al., 2020). Let $q_{(1)}^S \leq \dots \leq q_{(B-1)}^S$ and $q_{(1)}^T \leq \dots \leq q_{(B-1)}^T$ denote the sorted death times of the student embeddings $\{\mathbf{z}_i : i \in \mathcal{B}\}$ and the original teacher embeddings $\{\mathbf{t}_i : i \in \mathcal{B}\}$. We define

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left(q_{(j)}^S - q_{(j)}^T \right)^2. \quad (9)$$

For a precise interpretation, define the empirical distributions of finite connectivity scales

$$\nu_S = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{q_{(j)}^S}, \quad \nu_T = \frac{1}{B-1} \sum_{j=1}^{B-1} \delta_{q_{(j)}^T}, \quad (10)$$

where δ_a denotes a Dirac mass at a .

Proposition 3 (Transport interpretation and stability). *The H_0 loss equals the squared one-dimensional Wasserstein distance between the empirical death-scale measures,*

$$\mathcal{L}_{H_0} = W_2^2(\nu_S, \nu_T). \quad (11)$$

Moreover, if the two distance matrices on the corresponding batch satisfy

$$\epsilon = \max_{i,j \in \mathcal{B}} |d(\mathbf{z}_i, \mathbf{z}_j) - d(\mathbf{t}_i, \mathbf{t}_j)|, \quad (12)$$

then $\mathcal{L}_{H_0} \leq \epsilon^2$.

The proof is given in Appendix A.

The teacher diagram is computed from $\mathbf{t}_i \in \mathbb{R}^{d_T}$ rather than $\mathbf{y}_i$ or $\boldsymbol{\tau}_i^{(e)}$. The loss therefore needs neither a projection nor a shared orientation. It is differentiable almost everywhere away from distance ties and changes in the active minimum spanning tree (Hofer et al., 2019). In implementation, the discrete tree selection and sorting permutation are treated as fixed during backpropagation, while gradients flow through the selected student edge weights. The stability result is one-way. Equation 9 matches the multiset of H_0 death times but does not impose correspondence between individual minimum-spanning-tree edges, recover the full metric geometry, or preserve higher-dimensional topological features.

Algorithm 1 makes the nonsmooth implementation explicit. The teacher branch is evaluated without gradients. On the student branch, the MST edge selection and sorting permutation are detached, while the selected edge weights remain differentiable. Exact distance ties are passed to SciPy in row-major upper-triangular order. Although the selected active tree can depend on this library tie-break, every valid MST has the same sorted multiset of edge weights; the backward pass follows the particular active tree returned in that forward pass.

3.3 OBJECTIVE AND ALTERNATING OPTIMIZATION

The endpoint loss for the same training minibatch $\mathcal{B}$ is

$$\mathcal{L}_{\text{end}}^{(e)} = \frac{1}{B} \sum_{i \in \mathcal{B}} \left(1 - \langle \mathbf{z}_i, \boldsymbol{\tau}_i^{(e)} \rangle \right). \quad (13)$$

GATE-KD optimizes the two-term objective

$$\mathcal{L}^{(e)} = \mathcal{L}_{\text{end}}^{(e)} + \lambda_{\text{topo}} \mathcal{L}_{H_0}. \quad (14)$$

Algorithm 1 Native-space H_0 connectivity-scale loss for one minibatch

Require: Row-normalized teacher and student batches T_B and Z_B ; numerical floor ε .

```
1: for  $A \in \{T_B, Z_B\}$  do  
2:   Compute  $G \leftarrow AA^\top$  in FP32 and symmetrize  $G \leftarrow (G + G^\top)/2$ .  
3:   Compute  $D_{ij} \leftarrow \sqrt{\max(2 - 2G_{ij}, \varepsilon)}$  and set  $D_{ii} \leftarrow 0$ .  
4:   On a detached CPU float64 copy, add the same positive constant to every off-diagonal edge  
   and select an undirected MST with SciPy from the upper triangle.  
5:   Gather the selected weights from the unshifted  $D$  and sort them to obtain  $q_{(1:B-1)}^A$ .  
6: end for  
7: return  $(B-1)^{-1} \sum_{j=1}^{B-1} (q_{(j)}^Z - q_{(j)}^T)^2$ .
```

Here, $\mathcal{L}_{H_0}$ is evaluated on the same minibatch using the student embeddings and the corresponding unprojected teacher embeddings.

Optimization alternates between stochastic updates of the student parameters and exact updates of the nuisance coordinate gauge. It can be viewed as an inexact block-coordinate procedure over θ and $R \in O(d_S)$: with the student fixed, Equation 3 solves the endpoint term exactly over R on $\mathcal{C}$; with the gauge fixed, stochastic minibatch updates modify θ . During epoch e , $R^{(e)}$ is held fixed while the student is optimized with Equation 14. After epoch e , unless it is the final epoch, the updated student re-encodes $\mathcal{C}$ and Equation 3 produces $R^{(e+1)}$ for the next epoch. The H_0 term is unaffected by these gauge updates because it is computed from the native teacher embeddings.

Teacher embeddings and the PCA subspace are computed once before student optimization. Ordinary training steps use the cached teacher embeddings and require no online teacher forward pass. Each gauge update adds one student pass over the gauge-fit set and an SVD of a $d_S \times d_S$ matrix. This joint view does not by itself imply that repeated refitting improves generalization; the controlled analysis in Section 5.1 treats it as a secondary refinement beyond the initially fitted gauge.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Teacher–student configurations. We adopt the three teacher–student pairs evaluated by TALAS (Dao et al., 2026). They comprise Qwen3-Embedding-4B $\rightarrow$ BERT-base, BGE-M3 $\rightarrow$ MiniLMv2-L6-H768, and Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-L6-H384 (Devlin et al., 2019; Wang et al., 2021; Chen et al., 2024; Zhang et al., 2025). These settings span students from 22M to 109M parameters and embedding bottlenecks of 2560 $\rightarrow$ 768, 1024 $\rightarrow$ 768, and 1024 $\rightarrow$ 384, respectively. The Qwen teachers use last-token pooling, whereas BGE-M3 uses CLS pooling.

Distillation corpus. Following the data-construction protocol of TALAS (Dao et al., 2026), we form a shared distillation corpus by sampling approximately 5K unlabeled training texts from each of Emotion, WiC, and STS-B. We retain this protocol to enable a controlled, like-for-like comparison with TALAS and the reproduced baselines. No labels or task annotations are used during distillation, and every method uses the same corpus. We refer to these three benchmarks as source tasks and the remaining six as non-source tasks. Tables retain the compact labels IOD and OOD, respectively; here OOD means only that a benchmark did not supply distillation text, not an unrestricted distribution shift or corpus-disjoint pretraining regime.

Baselines. We compare GATE-KD with methods covering complementary transfer mechanisms. RKD matches inter-example relations (Park et al., 2019), while Stella uses a multi-stage embedding-distillation recipe (Zhang et al., 2024a). CDM and DSKD address cross-tokenizer alignment and dual-space transfer, respectively (Chen et al., 2025; Zhang et al., 2024b). EMO combines token-level relations with hidden-state alignment (Truong et al., 2025), and TALAS combines teacher anchoring with hierarchy-based self-distillation (Dao et al., 2026). Method-specific training hyperparameters are reported in Appendix B.

Table 1: Results across nine sentence-level benchmarks. Distilled-student scores are mean $\pm$ sample standard deviation over three seeds; best and second-best results are bold and underlined.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Averages		
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	IOD	OOD	All
(a) <i>Qwen3-Embedding-4B</i> $\rightarrow$ <i>BERT-base</i> (109M)												
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	74.25	84.48	81.07
Student base	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	42.43	60.08	54.20
RKD	89.76 $\pm$ 0.11	70.90 $\pm$ 0.31	59.98 $\pm$ 0.05	81.94 $\pm$ 0.26	75.59 $\pm$ 0.13	69.02 $\pm$ 0.18	72.93 $\pm$ 0.07	67.84 $\pm$ 0.18	73.72 $\pm$ 0.08	67.57 $\pm$ 0.05	76.49 $\pm$ 0.07	73.52 $\pm$ 0.06
Stella	88.48 $\pm$ 0.30	68.94 $\pm$ 0.17	53.69 $\pm$ 0.50	83.95 $\pm$ 0.10	75.39 $\pm$ 0.54	71.93 $\pm$ 0.09	69.87 $\pm$ 0.64	64.39 $\pm$ 0.21	70.86 $\pm$ 0.36	65.49 $\pm$ 0.17	75.17 $\pm$ 0.13	71.95 $\pm$ 0.06
CDM	88.13 $\pm$ 0.23	69.03 $\pm$ 0.20	52.83 $\pm$ 0.50	83.00 $\pm$ 0.06	77.10 $\pm$ 0.27	71.84 $\pm$ 0.09	67.89 $\pm$ 0.37	61.06 $\pm$ 0.37	69.19 $\pm$ 0.42	64.62 $\pm$ 0.28	74.37 $\pm$ 0.12	71.12 $\pm$ 0.18
DSKD	87.97 $\pm$ 0.16	68.90 $\pm$ 0.28	53.07 $\pm$ 0.31	83.03 $\pm$ 0.15	76.98 $\pm$ 0.59	71.86 $\pm$ 0.08	67.77 $\pm$ 0.66	61.11 $\pm$ 0.26	69.55 $\pm$ 0.12	64.83 $\pm$ 0.12	74.29 $\pm$ 0.14	71.14 $\pm$ 0.10
EMO	86.89 $\pm$ 0.07	68.99 $\pm$ 0.11	52.02 $\pm$ 1.09	79.74 $\pm$ 0.12	76.37 $\pm$ 0.35	69.62 $\pm$ 0.16	67.89 $\pm$ 0.09	56.46 $\pm$ 0.47	64.54 $\pm$ 0.50	62.06 $\pm$ 0.49	72.72 $\pm$ 0.15	69.17 $\pm$ 0.24
TALAS	90.53 $\pm$ 0.09	<u>73.77</u> $\pm$ 0.21	<u>68.47</u> $\pm$ 0.34	85.38 $\pm$ 0.05	81.83 $\pm$ 0.21	69.59 $\pm$ 0.14	<u>76.79</u> $\pm$ 0.06	<u>70.79</u> $\pm$ 0.19	78.50 $\pm$ 0.08	72.19 $\pm$ 0.07	<u>79.85</u> $\pm$ 0.07	<u>77.29</u> $\pm$ 0.04
GATE-KD	90.60 $\pm$ 0.21	73.86 $\pm$ 0.10	69.48 $\pm$ 0.22	<u>85.31</u> $\pm$ 0.11	<u>80.23</u> $\pm$ 0.13	68.19 $\pm$ 0.12	77.80 $\pm$ 0.14	75.15 $\pm$ 0.07	<u>78.44</u> $\pm$ 0.16	<u>72.04</u> $\pm$ 0.08	80.49 $\pm$ 0.03	77.67 $\pm$ 0.03
(b) <i>BGE-M3</i> $\rightarrow$ <i>MiniLMv2-H768</i> (66M)												
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	71.78	83.81	79.80
Student base	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	44.62	62.24	56.36
RKD	89.14 $\pm$ 0.09	73.29 $\pm$ 0.09	<u>62.23</u> $\pm$ 0.27	84.35 $\pm$ 0.06	78.74 $\pm$ 0.11	65.02 $\pm$ 0.10	73.20 $\pm$ 0.19	65.02 $\pm$ 0.29	71.97 $\pm$ 0.38	66.41 $\pm$ 0.07	77.29 $\pm$ 0.06	73.66 $\pm$ 0.04
Stella	86.82 $\pm$ 0.28	70.06 $\pm$ 0.12	56.61 $\pm$ 0.74	83.02 $\pm$ 0.12	74.63 $\pm$ 0.12	69.72 $\pm$ 0.05	69.08 $\pm$ 0.20	63.14 $\pm$ 0.28	67.10 $\pm$ 0.10	64.47 $\pm$ 0.27	74.46 $\pm$ 0.05	71.13 $\pm$ 0.11
CDM	87.82 $\pm$ 0.19	69.99 $\pm$ 0.09	58.32 $\pm$ 0.51	83.44 $\pm$ 0.07	75.56 $\pm$ 0.40	69.70 $\pm$ 0.02	65.93 $\pm$ 0.16	60.84 $\pm$ 0.34	65.31 $\pm$ 0.15	64.45 $\pm$ 0.13	73.93 $\pm$ 0.06	70.77 $\pm$ 0.02
DSKD	87.70 $\pm$ 0.16	70.35 $\pm$ 0.32	59.71 $\pm$ 0.61	82.99 $\pm$ 0.18	75.58 $\pm$ 0.12	69.59 $\pm$ 0.04	65.36 $\pm$ 0.37	61.21 $\pm$ 0.31	65.02 $\pm$ 0.18	64.77 $\pm$ 0.27	73.89 $\pm$ 0.10	70.83 $\pm$ 0.11
EMO	87.54 $\pm$ 0.18	71.03 $\pm$ 0.14	58.52 $\pm$ 0.30	82.46 $\pm$ 0.18	74.73 $\pm$ 0.58	69.20 $\pm$ 0.22	64.71 $\pm$ 0.55	56.89 $\pm$ 0.27	61.25 $\pm$ 0.03	62.99 $\pm$ 0.19	72.89 $\pm$ 0.10	69.59 $\pm$ 0.12
TALAS	89.83 $\pm$ 0.09	74.70 $\pm$ 0.09	61.65 $\pm$ 0.23	86.62 $\pm$ 0.05	<u>80.85</u> $\pm$ 0.22	66.72 $\pm$ 0.07	<u>75.40</u> $\pm$ 0.10	<u>67.08</u> $\pm$ 0.31	<u>77.08</u> $\pm$ 0.01	<u>68.48</u> $\pm$ 0.07	<u>79.08</u> $\pm$ 0.08	<u>75.55</u> $\pm$ 0.06
GATE-KD	90.17 $\pm$ 0.05	<u>73.74</u> $\pm$ 0.24	64.83 $\pm$ 0.72	<u>86.42</u> $\pm$ 0.09	80.97 $\pm$ 0.08	65.47 $\pm$ 0.09	76.77 $\pm$ 0.03	68.27 $\pm$ 0.04	77.63 $\pm$ 0.12	69.31 $\pm$ 0.21	79.39 $\pm$ 0.06	76.03 $\pm$ 0.11
(c) <i>Qwen3-Embedding-0.6B</i> $\rightarrow$ <i>MiniLMv2-H384</i> (22M)												
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	72.41	82.50	79.14
Student base	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	41.52	58.16	52.62
RKD	86.76 $\pm$ 0.07	69.87 $\pm$ 0.21	56.31 $\pm$ 0.35	81.70 $\pm$ 0.03	76.03 $\pm$ 0.20	68.56 $\pm$ 0.22	69.21 $\pm$ 0.18	64.26 $\pm$ 0.19	70.25 $\pm$ 0.21	65.04 $\pm$ 0.20	74.64 $\pm$ 0.09	71.44 $\pm$ 0.10
Stella	84.04 $\pm$ 0.04	68.73 $\pm$ 0.44	52.41 $\pm$ 0.28	81.50 $\pm$ 0.12	72.26 $\pm$ 0.32	68.65 $\pm$ 0.16	64.64 $\pm$ 0.19	60.59 $\pm$ 0.12	64.27 $\pm$ 0.20	61.78 $\pm$ 0.13	71.96 $\pm$ 0.13	68.57 $\pm$ 0.10
CDM	84.74 $\pm$ 0.21	69.52 $\pm$ 0.20	53.76 $\pm$ 0.95	81.15 $\pm$ 0.14	72.86 $\pm$ 0.30	68.29 $\pm$ 0.11	63.98 $\pm$ 0.36	60.43 $\pm$ 0.22	63.52 $\pm$ 0.24	61.86 $\pm$ 0.43	72.11 $\pm$ 0.19	68.69 $\pm$ 0.27
DSKD	84.76 $\pm$ 0.10	69.56 $\pm$ 0.20	53.14 $\pm$ 0.29	81.15 $\pm$ 0.01	72.81 $\pm$ 0.09	68.33 $\pm$ 0.04	63.78 $\pm$ 0.06	60.24 $\pm$ 0.19	63.53 $\pm$ 0.18	61.67 $\pm$ 0.03	72.05 $\pm$ 0.06	68.59 $\pm$ 0.04
EMO	83.54 $\pm$ 0.31	69.31 $\pm$ 0.06	54.48 $\pm$ 0.27	80.87 $\pm$ 0.13	72.00 $\pm$ 0.16	67.89 $\pm$ 0.09	62.19 $\pm$ 0.16	57.04 $\pm$ 0.17	61.24 $\pm$ 0.23	61.20 $\pm$ 0.09	70.82 $\pm$ 0.07	67.62 $\pm$ 0.02
TALAS	86.39 $\pm$ 0.13	<u>71.81</u> $\pm$ 0.11	<u>58.40</u> $\pm$ 0.30	84.25 $\pm$ 0.03	<u>79.29</u> $\pm$ 0.11	66.82 $\pm$ 0.02	<u>70.42</u> $\pm$ 0.10	<u>68.08</u> $\pm$ 0.22	<u>73.34</u> $\pm$ 0.10	<u>66.19</u> $\pm$ 0.11	<u>76.71</u> $\pm$ 0.04	<u>73.20</u> $\pm$ 0.06
GATE-KD	87.65 $\pm$ 0.06	72.32 $\pm$ 0.24	63.04 $\pm$ 0.28	<u>84.20</u> $\pm$ 0.06	81.60 $\pm$ 0.05	68.07 $\pm$ 0.12	72.48 $\pm$ 0.08	68.96 $\pm$ 0.18	74.76 $\pm$ 0.14	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06

Evaluation protocol. We evaluate frozen student embeddings on nine benchmarks from three task families. For classification on Banking77, Emotion, and Tweet, we fit a logistic-regression probe on each benchmark’s training split and report macro-F1. For pair classification on MRPC, SciTail, and WiC, we map cosine similarity to $[0, 1]$ and report average precision. For semantic textual similarity on SICK, STS12, and STS-B, we report Spearman correlation with the gold scores. All metrics are multiplied by 100. Avg. is the unweighted mean of the nine primary scores, while Avg. IOD and Avg. OOD average the three in-domain and six out-of-domain benchmarks, respectively. We report means and sample standard deviations over three seeds.

4.2 MAIN RESULTS

Table 1 compares GATE-KD with six distillation baselines across nine benchmarks and three teacher–student configurations. GATE-KD obtains the numerically highest overall average in all three configurations, improving over the next-best baseline by 0.38, 0.48, and 1.59 points in configurations (a), (b), and (c), respectively. The largest margin occurs in the evaluated configuration with the compact 22M-parameter student.

The task-level results show that this advantage is broad but not uniform. GATE-KD ranks first on 18 of the 27 individual benchmark comparisons, with its most consistent results on classification and semantic textual similarity. Other methods remain stronger on some pair-classification benchmarks, including MRPC and WiC. Relative to TALAS, GATE-KD improves Avg. OOD by 0.64, 0.31, and 1.16 points across configurations (a), (b), and (c), respectively. In configuration (a), it essentially matches TALAS on the source benchmarks, with 72.04 versus 72.19 Avg. IOD, so its aggregate gain there is driven by the six non-source benchmarks. In configurations (b) and (c), GATE-KD improves both Avg. IOD and Avg. OOD, with the largest gains for the compact 22M student.

4.3 STUDENT OPTIMIZATION EFFICIENCY

Table 2 isolates the repeated minibatch optimization path, reporting update latency, throughput, cumulative optimization time, and peak allocated memory. Exact measurement scope is given in Appendix B.

Table 2: Measured student-optimization cost on NVIDIA H200 GPUs. Values are mean $\pm$ sample standard deviation over three seeds; best and second-best values are bold and underlined. Timings isolate minibatch updates; exact scope is given in Appendix B.

Method	Avg. $\uparrow$	ms/step $\downarrow$	samples/s $\uparrow$	Opt. min $\downarrow$	peak GiB $\downarrow$
<i>(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)</i>					
RKD	73.52 $\pm$ 0.06	119.7 $\pm$ 18.3	<u>1081.0</u> $\pm$ 179.8	<u>1.20</u> $\pm$ 0.18	<u>1.72</u> $\pm$ 0.00
Stella	71.95 $\pm$ 0.06	130.7 $\pm$ 8.1	975.9 $\pm$ 58.8	1.27 $\pm$ 0.08	1.78 $\pm$ 0.01
CDM	71.12 $\pm$ 0.18	340.4 $\pm$ 13.6	374.2 $\pm$ 15.3	3.38 $\pm$ 0.12	9.31 $\pm$ 0.01
DSKD	71.14 $\pm$ 0.10	290.9 $\pm$ 9.0	437.7 $\pm$ 13.3	2.88 $\pm$ 0.09	9.45 $\pm$ 0.00
EMO	69.17 $\pm$ 0.24	721.1 $\pm$ 12.9	176.5 $\pm$ 3.1	7.02 $\pm$ 0.12	9.30 $\pm$ 0.00
TALAS	<u>77.29</u> $\pm$ 0.04	329.0 $\pm$ 12.5	387.1 $\pm$ 15.0	3.23 $\pm$ 0.11	1.43 $\pm$ 0.00
GATE-KD	77.67 $\pm$ 0.03	83.4 $\pm$ 3.8	1527.2 $\pm$ 70.8	0.84 $\pm$ 0.04	<u>1.72</u> $\pm$ 0.00
<i>(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)</i>					
RKD	73.66 $\pm$ 0.04	77.5 $\pm$ 8.8	<u>1656.4</u> $\pm$ 201.2	<u>0.79</u> $\pm$ 0.09	<u>1.07</u> $\pm$ 0.00
Stella	71.13 $\pm$ 0.11	152.8 $\pm$ 30.8	859.1 $\pm$ 195.6	1.49 $\pm$ 0.30	1.09 $\pm$ 0.00
CDM	70.77 $\pm$ 0.02	127.5 $\pm$ 13.8	1006.4 $\pm$ 115.8	1.30 $\pm$ 0.12	2.15 $\pm$ 0.01
DSKD	70.83 $\pm$ 0.11	115.0 $\pm$ 21.9	1137.2 $\pm$ 241.0	1.14 $\pm$ 0.20	2.22 $\pm$ 0.01
EMO	69.59 $\pm$ 0.12	394.6 $\pm$ 33.4	324.0 $\pm$ 28.3	3.87 $\pm$ 0.35	2.15 $\pm$ 0.01
TALAS	<u>75.55</u> $\pm$ 0.06	184.4 $\pm$ 4.6	690.2 $\pm$ 17.3	1.82 $\pm$ 0.03	0.85 $\pm$ 0.00
GATE-KD	76.03 $\pm$ 0.11	56.5 $\pm$ 7.7	2281.6 $\pm$ 310.0	0.57 $\pm$ 0.07	<u>1.07</u> $\pm$ 0.00
<i>(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)</i>					
RKD	71.44 $\pm$ 0.10	64.8 $\pm$ 13.7	<u>2012.3</u> $\pm$ 488.6	<u>0.68</u> $\pm$ 0.14	<u>0.41</u> $\pm$ 0.00
Stella	68.57 $\pm$ 0.10	150.7 $\pm$ 15.3	850.0 $\pm$ 81.5	1.47 $\pm$ 0.13	0.42 $\pm$ 0.00
CDM	68.69 $\pm$ 0.27	161.8 $\pm$ 21.6	796.3 $\pm$ 113.1	1.65 $\pm$ 0.20	1.53 $\pm$ 0.00
DSKD	68.59 $\pm$ 0.04	175.4 $\pm$ 8.2	726.4 $\pm$ 33.5	1.78 $\pm$ 0.06	1.56 $\pm$ 0.00
EMO	67.62 $\pm$ 0.02	499.3 $\pm$ 1.4	254.8 $\pm$ 0.7	4.90 $\pm$ 0.01	1.53 $\pm$ 0.00
TALAS	<u>73.20</u> $\pm$ 0.06	169.8 $\pm$ 15.4	753.7 $\pm$ 71.4	1.69 $\pm$ 0.14	0.34 $\pm$ 0.00
GATE-KD	74.79 $\pm$ 0.06	62.9 $\pm$ 3.1	2025.2 $\pm$ 98.3	0.64 $\pm$ 0.04	<u>0.41</u> $\pm$ 0.00

GATE-KD records the lowest mean update time and cumulative optimization time, and the highest throughput, among the evaluated methods in all three configurations. It does so while attaining the highest aggregate score in each configuration. The comparison characterizes the repeatedly executed optimization loop rather than end-to-end training or deployment cost.

5 WHAT DRIVES GATE-KD?

We organize the analysis around two empirical observations. First, within the evaluated controls, the endpoint interface obtains most of its numerical gain after establishing fitted coordinate comparability; the particular fixed subspace and repeated gauge updates are secondary design choices. Second, native-space H_0 provides a smaller complementary signal rather than replacing sample-wise endpoint supervision. Unless stated otherwise, controls use configuration (c), whose 384-dimensional student has the strongest width bottleneck. The variants come from matched rerun suites; their GATE-KD reference rows repeat configuration (c) from the main sweep for consistency. Extended controls are grouped by claim in Appendices C and D.

5.1 FITTED COORDINATE COMPARABILITY EXPLAINS MOST OF THE ENDPOINT GAIN

Table 3 holds the PCA subspace and full GATE-KD objective (endpoint supervision plus native-space H_0) fixed while varying how the endpoint coordinates are treated. The unaligned row trains directly against the fixed PCA coordinates and therefore doubles as the matched PCA-only endpoint baseline, the sentence-embedding analogue of the TCS target construction (Zhou et al., 2025). In configuration (c), no alignment, a random orientation, a fixed fitted gauge, one refit, and epoch-wise Procrustes reach 72.51, 73.02, 74.63, 74.64, and 74.79 Avg., respectively. The additional invariant control directly minimizes the endpoint loss over an orthogonal rotation on every minibatch and reaches 72.74 ± 0.07 . It improves only marginally over the unaligned controls and trails epoch-wise

Procrustes by 2.05 points. Separately, within the gauge-schedule controls, the fixed fitted gauge gains 2.12 points over no alignment and retains all but 0.16 points of the epoch-wise result. Taken together, these results argue against explaining the gain by merely weakening pointwise matching through a student-adaptive rotation. They are instead consistent with the benefit coming from one fitted orientation, shared across minibatches, that is consistent with the student’s initial coordinates. We note what these controls do not isolate: because $R^{(0)}$ is fitted to the initialized student, the schedule controls cannot separate a resolved cross-model coordinate mismatch from a target orientation that merely makes early optimization easier for that particular initialization. A control that fits $R^{(0)}$ on one initialization and trains independently initialized students against that fixed orientation would separate the two readings; we leave it to future work and interpret the gauge here as an empirically useful optimization device rather than as a transferable coordinate system.

Table 3: **Gauge mechanism under the full GATE-KD objective.** Every row uses the same PCA subspace, student, corpus, optimizer, and training schedule, including the same H_0 term. Fixed-target rows use the cosine endpoint loss against the indicated orientation; the invariant row instead minimizes that loss over an orthogonal rotation on each minibatch. The *None* row trains against the raw PCA coordinates and serves as the matched fixed-PCA endpoint baseline (a TCS-style target without a fitted orientation). Values are mean $\pm$ sample standard deviation over three seeds.

Orientation treatment	Updates	Config. (c) Avg. $\uparrow$
None (fixed PCA target)	0	72.51 ± 0.05
Random orthogonal	0	73.02 ± 0.05
Per-minibatch invariant	Every step	72.74 ± 0.07
Fixed $R^{(0)}$	0	74.63 ± 0.07
One refit	1	74.64 ± 0.06
Epoch-wise Procrustes	4	74.79 ± 0.06

Within these controls, most of the observed benefit therefore accompanies an initially fitted gauge: the one-refit and fixed-gauge means differ by only 0.01 points, while epoch-wise refitting adds a modest 0.16 points over the fixed-gauge result. We interpret refitting as a secondary optimization refinement, not as an independently established generalization mechanism. The invariant control uses batches of 128 in a 384-dimensional student space, so its rotation is underdetermined outside the current batch span. It tests the reviewed local invariant relaxation but does not rule out every possible global rotation-invariant objective. Matched subspace controls further show that aligned random projections remain within 0.13–0.34 points of PCA despite retaining substantially less teacher energy, while the updated uncentered-SVD run is 0.19 points below PCA. Within these controls, the proximity of the subspace constructions is more consistent with fitted coordinate comparability than with retained variance or a uniquely optimal PCA basis being the primary driver. Comparisons with learned endpoint maps, subspace-construction controls, target-rank sensitivity, and gauge-fit-set size appear in Appendix C.

5.2 H_0 PROVIDES A COMPLEMENTARY CONNECTIVITY-SCALE SIGNAL

In the structural-control comparison, endpoint-only supervision reaches 74.26 Avg., while native-space H_0 at $\lambda_{\text{topo}} = 0.5$ reaches 74.79, a 0.53-point refinement. Table 4 compares it with pairwise, local-scale, teacher-MST, and Gram objectives while holding the endpoint interface and optimization protocol fixed. The “None” row repeats the endpoint-only reference from Table 9, while the H_0 row repeats configuration (c) from the main comparison.

“None” denotes endpoint-only supervision. The alternatives test a sorted spanning path, directed nearest-neighbor distances, teacher-MST edge correspondence, and dense native-space Gram matching. The H_0 objective instead matches the sorted MST edge-weight distributions without edge correspondence. Appendix D.2 gives the complete definitions and constraint counts.

Table 4: **Matched structural controls in configuration (c).** All variants retain the same PCA-Procrustes endpoint objective and differ only in the auxiliary structural statistic.

Auxiliary term	Avg. $\uparrow$	Δ endpoint
None	74.26 $\pm$ 0.02	—
Sorted spanning path	73.66 $\pm$ 0.04	−0.60
kNN distribution	74.47 $\pm$ 0.01	+0.21
Teacher-MST edges	74.33 $\pm$ 0.08	+0.07
Native Gram matching	73.57 $\pm$ 0.01	−0.69
H_0	74.79$\pm$ 0.06	+0.53

Among the evaluated controls and weights, the H_0 objective is the strongest auxiliary, exceeding kNN distribution matching by 0.32 points and teacher-MST matching by 0.46 points; sorted spanning-path and native Gram matching instead reduce Avg. by 0.60 and 0.69 points. Thus the gain is not explained by adding an arbitrary structural objective and is consistent with connectivity-scale matching providing a distinct complementary signal under the evaluated weights. Appendix D.1 shows that H_0 alone reaches 64.22, so connectivity does not replace pointwise correspondence. Topology-weight sensitivity appears in Appendix D.3.

6 CONCLUSION

GATE-KD treats cross-dimensional embedding distillation as an interface problem, using a fixed PCA subspace and closed-form gauge alignment with native-space H_0 regularization. Across three teacher–student configurations, this final-embedding-only formulation obtains the numerically highest aggregate scores among the evaluated distilled students while recording the lowest update time and cumulative optimization time within the measured minibatch loop. Controlled analyses show that fitted coordinate comparability provides the dominant gain: a per-minibatch rotation-invariant control trails GATE-KD by 2.05 points, while separate schedule controls identify repeated refitting as a secondary optimization refinement with a smaller numerical increment. Native-space H_0 adds 0.53 points in the compact comparison and performs best among the evaluated structural auxiliaries. The current sentence-level objective does not fully preserve the teacher’s zero-shot retrieval capability, leaving broader retrieval transfer for future work. In addition, the current corpus uses training text from three downstream source benchmarks, so its source-task results measure label-free task adaptation rather than corpus-disjoint transfer. The conclusions are therefore limited to the task-adapted sentence-level regime evaluated here.

REFERENCES

- Mohammad Kalim Akram, Saba Sturua, Nastia Havriushenko, Quentin Herreros, Michael Günther, Maximilian Werk, and Han Xiao. jina-embeddings-v5-text: Task-targeted embedding distillation. arXiv:2602.15547, 2026. URL <https://arxiv.org/abs/2602.15547>.
- Prajwal Bhattarai, Mohammad Amjad, Dmytro Zhylo, and Tuka Alhanai. Knowledge distillation through geometry-aware representational alignment. arXiv:2509.25253, 2025. URL <https://arxiv.org/abs/2509.25253>.
- A. I. Burykina, D. R. Ledneva, and D. P. Kuznetsov. JDCEMB: Joint distillation and contrastive learning for embeddings in task-oriented dialogue systems. *Doklady Mathematics*, 112:319–331, 2025. doi: 10.1134/S1064562425700243.
- Frédéric Chazal, Vin de Silva, and Steve Oudot. Persistence stability for geometric complexes. *Geometriae Dedicata*, 173:193–214, 2014. doi: 10.1007/s10711-013-9937-z. URL <https://arxiv.org/abs/1207.3885>.
- Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2318–2335, 2024. doi: 10.18653/v1/2024.findings-acl.137. URL <https://aclanthology.org/2024.findings-acl.137/>.

- Yijie Chen, Yijin Liu, Fandong Meng, Yufeng Chen, Jinan Xu, and Jie Zhou. Enhancing cross-tokenizer knowledge distillation with contextual dynamical mapping. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 8005–8018, 2025. doi: 10.18653/v1/2025.findings-acl.419. URL <https://aclanthology.org/2025.findings-acl.419/>.
- Yudong Chen, Sen Wang, Jiajun Liu, Xuwei Xu, Frank de Hoog, Brano Kusy, and Zi Huang. Understanding the effects of projectors in knowledge distillation. arXiv:2310.17183, 2023. URL <https://arxiv.org/abs/2310.17183>.
- Quoc Phong Dao, Hoang Son Nguyen, Khanh Chi Pham, Linh Ngo Van, Diep Thi-Ngoc Nguyen, Thien Huu Nguyen, and Trung Le. TALAS: Teacher-anchored layer alignment with adaptive sharpness-aware minimization for embedding distillation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 32705–32723, 2026. doi: 10.18653/v1/2026.acl-long.1509. URL <https://aclanthology.org/2026.acl-long.1509/>.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 4171–4186, 2019. doi: 10.18653/v1/N19-1423. URL <https://aclanthology.org/N19-1423/>.
- Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936. doi: 10.1007/BF02288367.
- Chaochen Gao, Xing Wu, Peng Wang, Jue Wang, Liangjun Zang, Zhongyuan Wang, and Songlin Hu. DistilCSE: Effective knowledge distillation for contrastive sentence embeddings. arXiv:2112.05638, 2021.
- Sang Il Han. Teacher supervision over representation equivalence classes. arXiv:2607.03572, 2026. URL <https://arxiv.org/abs/2607.03572>.
- Edith Heiter, Robin Vandaele, Tijl De Bie, Yvan Saeys, and Jefrey Lijffijt. Topologically regularized data embeddings. arXiv:2301.03338, 2023.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Christoph Hofer, Roland Kwitt, Marc Niethammer, and Mandar Dixit. Connectivity-optimized representation learning via persistent homology. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 2751–2760, 2019. URL <https://proceedings.mlr.press/v97/hofer19a.html>.
- Christoph Hofer, Florian Graf, Bastian Rieck, Marc Niethammer, and Roland Kwitt. Topologically densified distributions. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 4304–4313, 2020. URL <https://proceedings.mlr.press/v119/hofer20a.html>.
- [FILL FULL AUTHOR LIST] Huy. MIPIC: Matryoshka representation learning via self-distilled intra-relational and progressive information chaining, 2026. Preprint.
- Eun Som Jeon, Rahul Khurana, Aishani Pathak, and Pavan Turaga. Leveraging topological guidance for improved knowledge distillation. In *Proceedings of the Geometry-grounded Representation Learning and Generative Modeling Workshop*, volume 251 of *Proceedings of Machine Learning Research*, pp. 158–172, 2024. URL <https://proceedings.mlr.press/v251/jeon24a.html>.
- Chaitanya K. Joshi, Fayao Liu, Xu Xun, Jie Lin, and Chuan-Sheng Foo. On representation knowledge distillation for graph neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 35(4):4656–4667, 2024. doi: 10.1109/TNNLS.2022.3223018.

594 Jungeun Kim, Junwon You, Dongjin Lee, Ha Young Kim, and Jae-Hun Jung. Do topological char-
595 acteristics help in knowledge distillation? In *Proceedings of the 41st International Conference on*
596 *Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 24674–24693,
597 2024. URL <https://proceedings.mlr.press/v235/kim24aj.html>.

598
599 Seungyeon Kim, Ankit Singh Rawat, Manzil Zaheer, Sadeep Jayasumana, Veeranjaneyulu Sad-
600 hanala, Wittawat Jitkrittum, Aditya Krishna Menon, Rob Fergus, and Sanjiv Kumar. EmbedDis-
601 till: A geometric knowledge distillation for information retrieval. arXiv:2301.12005, 2023. URL
602 <https://arxiv.org/abs/2301.12005>.

603
604 Jongwoo Ko, Seungjoon Park, Minchan Jeong, Sukjin Hong, Euijai Ahn, Du-Seong Chang, and
605 Se-Young Yun. Revisiting intermediate layer distillation for compressing language models: An
606 overfitting perspective. In *Findings of the Association for Computational Linguistics: EACL 2023*,
607 2023.

608
609 Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural
610 network representations revisited. In *International Conference on Machine Learning*, 2019.

611
612 Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, Vivek Ra-
613 manujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi.
614 Matryoshka representation learning. In *Advances in Neural Information Processing Systems*,
615 2022.

616
617 Jinhyuk Lee, Zhuyun Dai, Xiaoqi Ren, Blair Chen, Daniel Cer, Jeremy R. Cole, Kai Hui, Michael
618 Boratko, Rajvi Kapadia, Wen Ding, Yi Luan, Sai Meher Karthik Duddu, Gustavo Hernandez
619 Abrego, Weiqiang Shi, Nithi Gupta, Aditya Kusupati, Prateek Jain, Siddhartha Reddy Jonnal-
620 agadda, Ming-Wei Chang, and Iftexhar Naim. Gecko: Versatile text embeddings distilled from
621 large language models. arXiv:2403.20327, 2024. URL <https://arxiv.org/abs/2403.20327>.

622
623 Xianming Li, Zongxi Li, Jing Li, Haoran Xie, and Qing Li. ESE: Espresso sentence embeddings.
624 In *International Conference on Learning Representations*, 2025.

625
626 Zhuoyuan Mao and Tetsuji Nakagawa. LEALLA: Learning lightweight language-agnostic sen-
627 tence embeddings with knowledge distillation. In *Proceedings of the 17th Conference of the*
628 *European Chapter of the Association for Computational Linguistics*, 2023. URL <https://aclanthology.org/2023.eacl-main.138/>.

629
630 Lucas Maystre, Alvaro Ortega Gonzalez, Charles Park, Rares Dolga, Tudor Berariu, Yu Zhao,
631 and Kamil Ciosek. When embedding models meet: Procrustes bounds and applications.
632 arXiv:2510.13406, 2025. URL <https://arxiv.org/abs/2510.13406>.

633
634 Roy Miles and Krystian Mikolajczyk. Understanding the role of the projector in knowledge distil-
635 lation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

636
637 Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. Relational knowledge distillation. In *Pro-*
638 *ceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.

639
640 Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends*
641 *in Machine Learning*, 11(5–6):355–607, 2019. doi: 10.1561/22000000073. URL <https://arxiv.org/abs/1803.00567>.

642
643 Nils Reimers and Iryna Gurevych. Making monolingual sentence embeddings multilingual using
644 knowledge distillation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural*
645 *Language Processing*, pp. 4512–4525, 2020.

646
647 Peter H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*,
31(1):1–10, 1966.

Yonglong Tian, Dilip Krishnan, and Phillip Isola. Contrastive representation distillation. In *Inter-
national Conference on Learning Representations*, 2020.

- Minh-Phuc Truong, Hai An Vu, Tu Vu, Diep Thi-Ngoc Nguyen, Linh Ngo Van, Thien Huu Nguyen, and Trung Le. EMO: Embedding model distillation via intra-model relation and optimal transport alignments. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025.
- Frederick Tung and Greg Mori. Similarity-preserving knowledge distillation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019.
- Hai An Vu, Minh-Phuc Truong, Tu Vu, and Linh Ngo Van. MoL: Mixture of layers in cross-tokenizer embedding model distillation. *Knowledge-Based Systems*, pp. 116001, 2026. doi: 10.1016/j.knosys.2026.116001.
- Robin Vujanic and Thomas Rückstieß. LEAF: Knowledge distillation of text embedding models with teacher-aligned representations. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 43362–43383, 2026. doi: 10.18653/v1/2026.acl-long.2008. URL <https://aclanthology.org/2026.acl-long.2008/>.
- Shuo Wang, Chihang Wang, Jia Gao, Zhen Qi, Hongye Zheng, and Xiaoxuan Liao. Feature alignment-based knowledge distillation for efficient compression of large language models. arXiv:2412.19449, 2024. URL <https://arxiv.org/abs/2412.19449>.
- Wenhui Wang, Hangbo Bao, Shaohan Huang, Li Dong, and Furu Wei. MiniLMv2: Multi-head self-attention relation distillation for compressing pretrained transformers. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pp. 2140–2151, 2021. doi: 10.18653/v1/2021.findings-acl.188. URL <https://aclanthology.org/2021.findings-acl.188/>.
- Alex H. Williams, Erin Kunz, Simon Kornblith, and Scott W. Linderman. Generalized shape metrics on neural representations. In *Advances in Neural Information Processing Systems*, 2021.
- Jian Xie, Xin He, Jiyang Wang, Zimeng Qiu, Ali Kebarighotbi, and Farhad Ghassemi. SimTDE: Simple transformer distillation for sentence embeddings. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 2389–2393, 2023. doi: 10.1145/3539618.3592063.
- Jiahao Xu, Wei Shao, Lihui Chen, and Lemao Liu. DistillCSE: Distilled contrastive learning for sentence embeddings. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 8153–8165, 2023.
- Zony Yu, Yuqiao Wen, and Lili Mou. Revisiting intermediate-layer matching in knowledge distillation: Layer-selection strategy doesn’t matter (much). In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pp. 1686–1694, 2025. doi: 10.18653/v1/2025.findings-ijcnlp.105. URL <https://aclanthology.org/2025.findings-ijcnlp.105/>.
- Dun Zhang, Jiacheng Li, Ziyang Zeng, and Fulong Wang. Jasper and stella: Distillation of sota embedding models. *arXiv preprint arXiv:2412.19048*, 2024a.
- Songming Zhang, Xue Zhang, Zengkui Sun, Yufeng Chen, and Jinan Xu. Dual-space knowledge distillation for large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 18164–18181, 2024b. doi: 10.18653/v1/2024.emnlp-main.1010. URL <https://aclanthology.org/2024.emnlp-main.1010/>.
- Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. Qwen3 Embedding: Advancing text embedding and reranking through foundation models. arXiv:2506.05176, 2025. URL <https://arxiv.org/abs/2506.05176>.
- Xuandong Zhao, Zhiguo Yu, Ming Wu, and Lei Li. Compressing sentence representation for semantic retrieval via homomorphic projective distillation. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022.

702 Junjie Zhou, Ke Zhu, and Jianxin Wu. All you need in knowledge distillation is a tailored coordinate
703 system. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
704

705 Zhenyu Zhou, Defang Chen, Can Wang, Chun Chen, and Siwei Lyu. DICE: Distilling classifier-free
706 guidance into text embeddings. In *Proceedings of the AAAI Conference on Artificial Intelligence*,
707 2026. URL <https://arxiv.org/abs/2502.03726>.
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A PROOFS OF SUPPORTING RESULTS

Proof of Proposition 1. For unit-norm rows, $1 - \langle \mathbf{z}_i^{(e)}, \mathbf{y}_i R \rangle = \frac{1}{2} \|\mathbf{z}_i^{(e)} - \mathbf{y}_i R\|_2^2$. Summing over $\mathcal{C}$ gives Equation 5. Expanding the Frobenius norm shows that minimizing it over $R \in O(d_S)$ is equivalent to maximizing $\text{tr}(R^\top Y_C^\top Z_C^{(e)})$. If $Y_C^\top Z_C^{(e)} = U \Sigma V^\top$, the orthogonal Procrustes solution is $R = UV^\top$, proving optimality. Finally, $(YR)(YR)^\top = YRR^\top Y^\top = YY^\top$ for every orthogonal R . The same global transformation is applied rowwise, so sentence–target correspondence is also unchanged. $\square$

Proof of Proposition 2. The eigenvalues of $TT^\top$ are $\sigma_j(T)^2$, whereas $\text{rank}(ZZ^\top) \leq d_S$. The Eckart–Young theorem therefore lower-bounds the squared error of any such rank- d_S approximation by the squared tail eigenvalues (Eckart & Young, 1936). $\square$

Proof of Proposition 3. Optimal transport on the real line pairs equally weighted samples in sorted order, which gives Equation 11 (Peyré & Cuturi, 2019). Under a uniform edge perturbation of at most ϵ , every threshold graph at scale a in either metric is contained in the other at scale $a + \epsilon$. Kruskal’s characterization of the ordered minimum-spanning-tree edges then gives $|q_{(j)}^S - q_{(j)}^T| \leq \epsilon$ for every j , and averaging their squared differences proves the bound. This is the degree-zero finite-space case of Vietoris–Rips persistence stability (Chazal et al., 2014). $\square$

B TRAINING DETAILS

All main-comparison runs use a maximum sequence length of 256 and seeds 42, 43, and 44. We fix the batch size at 128 and the total training duration at five epochs for every method; Stella’s two stages comprise two and three epochs. Learning rate is the only optimization hyperparameter selected separately for each method. Selection is performed on validation data before the three-seed test sweep; the selected rate is then held fixed across all three teacher–student configurations and seeds. The final-epoch checkpoint is evaluated once on the test splits. No test score is used for learning-rate or checkpoint selection. Table 5 reports every selected rate and the method-specific optimizer settings. For every method, the learning rate was tuned over the common grid $\{1, 2, 3, 5, 7\} \times 10^{-5}$; no other optimizer hyperparameter was tuned per method. All methods use AdamW with weight decay 0.01 and a cosine schedule ending at 2×10^{-6} . TALAS wraps AdamW in adaptive SAM. For GATE-KD, $\lambda_{\text{topo}} = 0.5$, the H_0 cloud is the current training batch, the gauge is refitted after each non-final epoch, and gauge fitting uses the full distillation corpus. These settings are fixed before test evaluation; the sensitivity sweeps are diagnostic rather than part of model selection. Table 5 lists the selected learning rates, optimizers, and warm-up ratios.

Table 5: Main-experiment training hyperparameters. Learning rate is the only optimization hyperparameter selected separately by method; each selected value is fixed across all three configurations and reported seeds.

Method	Epochs	Batch	Learning rate	Optimizer	Warm-up
RKD	5	128	7×10^{-5}	AdamW	0.06
Stella	2 + 3	128	5×10^{-5}	AdamW	0.06
CDM	5	128	2×10^{-5}	AdamW	0.06
DSKD	5	128	2×10^{-5}	AdamW	0.06
EMO	5	128	1×10^{-5}	AdamW	0.10
TALAS	5	128	2×10^{-5}	AdamW + adaptive SAM	0.06
GATE-KD	5	128	7×10^{-5}	AdamW	0.06

Optimization-timing scope. CUDA events measure each minibatch update, including the student forward pass, loss computation, backward pass, and optimizer update. Mean update time and throughput exclude the first ten warm-up steps; cumulative optimization time sums all measured updates across five epochs. One-time preparation, epoch-boundary gauge refitting, model and data loading, and final evaluation fall outside this timed region. The archived sweep includes co-located processes, so the reported values characterize the measured optimization environment rather than isolated-process or end-to-end efficiency.

C ENDPOINT-INTERFACE ANALYSIS

C.1 COMPARISON WITH LEARNED ENDPOINT MAPS

Table 6 complements the gauge-schedule analysis in Table 3 by varying the broader endpoint interface. All rows use endpoint-only supervision in configuration (c). The learned controls use a single bias-free linear map, either from teacher to student width or from student to teacher width, followed by row normalization. The map is trained jointly with the student under the same AdamW schedule and is discarded at inference. Thus these rows compare supervision interfaces rather than deployed model capacities.

Table 6: Endpoint-interface ablation with endpoint-only supervision in configuration (c). Scores are mean $\pm$ sample standard deviation over three seeds; random controls share one map draw. Best values are bold.

Target map	Gauge	Retained energy $\uparrow$	Avg. IOD $\uparrow$	Avg. OOD $\uparrow$	Avg. $\uparrow$
Learned $T \rightarrow S$	—	N/A	54.34 ± 0.68	65.02 ± 0.25	61.46 ± 0.39
Learned $S \rightarrow T$	—	N/A	62.54 ± 0.36	72.72 ± 0.31	69.33 ± 0.32
PCA	None	0.928	62.49 ± 0.19	72.78 ± 0.07	69.35 ± 0.04
PCA	Random orthogonal	0.928	63.68 ± 0.10	73.72 ± 0.09	70.37 ± 0.08
Random subspace	None	0.381	64.44 ± 0.14	74.16 ± 0.04	70.92 ± 0.07
PCA	Procrustes	0.928	67.76 ± 0.05	77.54 ± 0.07	74.28 ± 0.03

Under the matched endpoint-only recipe evaluated here, the fixed PCA–Procrustes interface exceeds learned linear maps in both directions. The unaligned PCA row is the endpoint-only counterpart of the fixed-PCA (TCS-style) target in Table 3; without a fitted gauge it trails the Procrustes row by 4.93 points and is within 1.6 points of a random subspace of the same rank. This comparison is specific to the stated single-layer maps and optimization protocol; it does not establish that every learned projector is inferior. Table 3 isolates the role of the fitted gauge, while Appendix C.2 holds that gauge fixed across subspace choices.

C.2 ROBUSTNESS TO FIXED-SUBSPACE CONSTRUCTION

We isolate teacher projection in configuration (c), holding the full GATE-KD objective (native-space H_0 , $\lambda_{\text{topo}} = 0.5$), student, optimizer, schedule, and epoch-wise Procrustes fixed. These matched reruns need not equal the main comparison numerically. The reference fits PCA directions on the centered cache but applies them without subtracting the mean; mean-subtracted PCA applies this affine centering, and whitening also rescales by inverse square root eigenvalues. Random controls use three independent Haar subspaces of the same rank. Each row averages seeds 42–44, with random draws kept separate.

Table 7: PCA preprocessing and subspace controls under the full GATE-KD objective in configuration (c). All rows retain native-space H_0 ($\lambda_{\text{topo}} = 0.5$) and epoch-wise Procrustes; only the teacher projection changes. Mean $\pm$ sample standard deviation over three seeds; random draws remain separate; best values are bold.

Interface	Retained energy $\uparrow$	Avg. IOD $\uparrow$	Avg. OOD $\uparrow$	Avg. $\uparrow$
PCA + Procrustes	0.928	68.62 ± 0.21	77.87 ± 0.02	74.79 ± 0.06
Mean-subtracted PCA	0.928	66.49 ± 0.12	74.89 ± 0.02	72.09 ± 0.05
PCA whitening	0.928	66.44 ± 0.14	74.16 ± 0.04	71.59 ± 0.06
Random subspace (draw 0)	0.381	68.05 ± 0.16	77.65 ± 0.08	74.45 ± 0.11
Random subspace (draw 1)	0.376	68.35 ± 0.28	77.82 ± 0.04	74.66 ± 0.07
Random subspace (draw 2)	0.374	68.58 ± 0.16	77.40 ± 0.05	74.46 ± 0.06
Uncentered SVD	0.945	68.67 ± 0.08	77.82 ± 0.03	74.60 ± 0.03

Mean subtraction lowers Avg. from 74.79 to 72.09 (-2.70), and whitening to 71.59 (-3.20), with both IOD and OOD declining. Aligned random subspaces reach 74.45–74.66, only 0.13–0.34 below

PCA despite retaining 0.374–0.381 rather than 0.928 of the teacher energy. The updated uncentered-SVD run reaches 74.60, 0.19 points below PCA overall. Thus preserving the mean component at application time is important, whereas retained energy and the exact fitted subspace appear less decisive once the gauge is aligned. PCA is numerically best here, but the proximity of both SVD and aligned random subspaces does not establish fit-time centering as uniquely optimal.

C.3 ENDPOINT-INTERFACE RANK AT FIXED STUDENT WIDTH

We isolate the subspace selected by the endpoint interface in configuration (c). The student architecture, 384-dimensional output, distillation corpus, optimizer, training schedule, and Procrustes updates remain fixed, while the retained PCA rank varies over $k \in \{64, 128, 256, 384\}$. This experiment uses the endpoint-only objective. Targets with $k < 384$ are embedded in the native student space by appending zero coordinates before Procrustes alignment. This padding preserves their Gram matrix and active rank, and does not change the capacity or width of the student.

Retained energy is measured on the complete teacher cache. On a fixed sample of 2,048 normalized teacher embeddings T , we measure the normalized squared Gram distortion $\|TT^\top - Y_k Y_k^\top\|_F^2 / \|TT^\top\|_F^2$, where Y_k is the normalized rank- k PCA target. We also report the normalized rank floor $\sum_{j>k} \sigma_j(T)^4 / \sum_j \sigma_j(T)^4$ from Equation 7. The Gram quantities are independent of the orthogonal gauge and are therefore computed once, while downstream scores are averaged over three seeds.

Figure 1 shows monotonic changes in all three quantities. Increasing k from 64 to 384 raises retained teacher energy from 59.32% to 92.84%, reduces normalized Gram distortion from 6.797% to 0.086%, and improves Avg. from 72.57 to 74.27. The intermediate ranks 128 and 256 attain 73.77 and 74.21 Avg., respectively. Most of the downstream benefit is therefore recovered by rank 256: it captures 96% of the 64-to-384 improvement, with the final increase to rank 384 contributing only 0.06 points in this configuration.

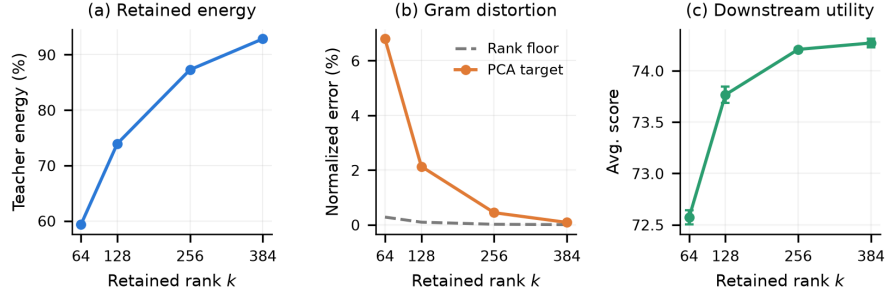


Figure 1: Sensitivity to retained PCA rank at fixed student width in configuration (c), endpoint-only objective, one corpus, four ranks. Targets with $k < 384$ are zero-padded before Procrustes alignment. Downstream scores show mean $\pm$ sample standard deviation over three seeds; retained energy and Gram distortion are gauge-independent and computed once.

PCA distortion remains roughly 21–25 times above the rank floor because its objective differs from normalized Gram approximation: PCA minimizes reconstruction error of the raw coordinates, whereas the Gram quantity compares inner products after re-normalizing each projected row, and re-normalization amplifies the truncated tail for rows with small retained norm. The floor therefore bounds irreducible error rather than predicting this interface’s distortion, and retained energy should not be read as a proxy for how well the teacher’s angular geometry is preserved; only the Gram distortion column measures that directly. Two further scope notes apply. First, because the padded target has zero coordinates outside its rotated k -dimensional subspace, the cosine endpoint loss drives the student’s remaining $384 - k$ output coordinates toward zero rather than leaving them free, so for $k < 384$ the student effectively operates at rank k despite its 384-dimensional output. Second, the sweep is endpoint-only, in configuration (c), on one corpus; we have not repeated it with the H_0 term or in configurations (a) and (b). Within these limits, increasing rank improves

geometry and utility with diminishing returns. With the student architecture fixed, this is target-rank sensitivity, not a width ablation.

C.4 GAUGE-FIT-SET SIZE

Table 8 varies only the number of corpus examples used to estimate the orthogonal gauge in configuration (c). Reducing the fit set from all 14,760 sentences to 2,048 changes Avg. by 0.28 points. This suggests that the fitted coordinate relation can be estimated from a modest sample under the evaluated protocol; it does not establish out-of-distribution generalization of the gauge.

Table 8: Gauge-fit-set-size sensitivity in configuration (c), reported as mean $\pm$ sample standard deviation over three seeds. Best values are bold.

Fit-set size	Avg. IOD $\uparrow$	Avg. OOD $\uparrow$	Avg. $\uparrow$
2,048	68.36 $\pm$ 0.11	77.59 $\pm$ 0.04	74.51 $\pm$ 0.06
4,096	68.52 $\pm$ 0.08	77.74 $\pm$ 0.05	74.67 $\pm$ 0.02
8,192	68.26 $\pm$ 0.08	77.78 $\pm$ 0.03	74.61 $\pm$ 0.02
14,760	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06

D STRUCTURAL-REGULARIZATION ANALYSIS

D.1 ENDPOINT AND TOPOLOGY SIGNAL DECOMPOSITION

Table 9 separates the endpoint and topology signals. The endpoint-only objective supplies most of the downstream gain, whereas H_0 -only training reduces the connectivity residual without reaching the endpoint-supervised score. Relative to the no-teacher control, endpoint-only training gains 21.66 points, compared with 11.62 points for H_0 alone. Adding native-space H_0 to the endpoint objective yields a further 0.53 points and reduces the H_0 residual from 0.2122 to 0.1281. The combined objective therefore treats native-space connectivity as an auxiliary constraint rather than as a substitute for sample-wise correspondence. The H_0 -only row uses $\lambda_{\text{topo}} = 1$, whereas the combined row repeats the $\lambda_{\text{topo}} = 0.5$ main reference; their difference should therefore be read as a signal decomposition rather than a matched weight ablation.

Table 9: Primary endpoint and topology signal decomposition in configuration (c). Values are mean $\pm$ sample standard deviation over three seeds; H_0 residuals use the native teacher space. Best values are bold.

Objective	λ_{topo}	H_0 residual $\downarrow$	Avg. IOD $\uparrow$	Avg. OOD $\uparrow$	Avg. $\uparrow$
No-teacher control	—	0.3595 $\pm$ 0.0000	41.61 $\pm$ 0.00	58.10 $\pm$ 0.00	52.60 $\pm$ 0.00
H_0 only	1.0	0.1240 $\pm$ 0.0012	58.00 $\pm$ 1.01	67.33 $\pm$ 0.62	64.22 $\pm$ 0.73
Endpoint only	0	0.2122 $\pm$ 0.0004	67.68 $\pm$ 0.19	77.55 $\pm$ 0.07	74.26 $\pm$ 0.02
Endpoint + native-space H_0	0.5	0.1281 $\pm$ 0.0003	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06

D.2 STRUCTURAL-CONTROL DEFINITIONS

The sorted spanning-path control constructs a geometry-independent path through the shuffled batch and matches the independently sorted student and teacher edge lengths. The kNN control similarly matches independently sorted directed nearest-neighbor distances with $k = 1$. Teacher-MST first selects the teacher’s $B - 1$ minimum-spanning-tree edges and then compares student distances on those same sample pairs, retaining edge correspondence. Native Gram matching compares all pair-wise cosine similarities. Finally, H_0 matches independently sorted student and teacher MST edge weights and discards edge correspondence. The path, Teacher-MST, and H_0 controls expose $B - 1$ scalar constraints per batch, kNN exposes B , and Gram matching is deliberately denser.

D.3 TOPOLOGY-WEIGHT SENSITIVITY

Table 10 varies the topology weight in configuration (c), with all other settings fixed to the main recipe. Every tested nonzero weight improves over the endpoint-only setting, while values from 0.5 to 1.0 differ by at most 0.24 points. We omit the earlier epoch-matched batch-size sweep because changing batch size also changed the number of optimizer updates and therefore did not isolate batch size.

Table 10: Topology-weight sensitivity in configuration (c), reported as mean $\pm$ sample standard deviation over three seeds. Best values are bold.

λ_{topo}	Avg. IOD $\uparrow$	Avg. OOD $\uparrow$	Avg. $\uparrow$
0	67.54 $\pm$ 0.18	77.54 $\pm$ 0.07	74.21 $\pm$ 0.03
0.5	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06
0.75	68.47 $\pm$ 0.13	77.84 $\pm$ 0.04	74.72 $\pm$ 0.04
1	68.14 $\pm$ 0.06	77.75 $\pm$ 0.03	74.55 $\pm$ 0.03

E ADDITIONAL EVALUATION

E.1 BROADER 100K TASK-ADAPTED CORPUS

We repeat the complete method sweep on approximately 100K unlabeled texts drawn from the training splits of all nine sentence-level benchmarks. No labels or task annotations are used. Relative to the main comparison, this setting changes both corpus size and dataset coverage. It therefore evaluates robustness in a broader task-adapted setting rather than isolating the effect of corpus size alone.

Table 11: Results with the broader 100K distillation corpus. Distilled-student scores are mean $\pm$ sample standard deviation over three seeds; best and second-best results are bold and underlined.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Avg.
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	
(a) Qwen3-Embedding-4B → BERT-base (109M)										
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	81.07
Student base	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	54.20
RKD	92.23±0.15	71.96±0.18	63.79±0.40	85.00±0.04	80.27±0.38	66.31±0.10	79.32±0.02	71.72±0.16	77.70±0.20	76.48±0.04
Stella	90.02±0.15	71.90±0.22	60.14±0.45	85.96±0.05	78.82±0.27	69.93±0.13	76.53±0.23	69.79±0.12	77.22±0.16	75.59±0.10
CDM	89.16±0.04	69.63±0.23	53.74±0.13	85.57±0.08	71.22±0.19	70.04±0.05	71.26±0.29	66.97±0.21	73.69±0.38	72.36±0.13
DSKD	88.87±0.07	69.05±0.09	54.02±0.56	85.90±0.05	70.02±0.31	69.63±0.21	71.07±0.10	66.52±0.17	73.52±0.01	72.07±0.03
EMO	84.96±0.84	68.93±0.65	54.80±1.52	85.05±0.04	65.75±0.22	68.80±0.13	62.00±2.71	61.75±1.85	69.54±0.37	69.07±0.89
TALAS	93.12±0.13	74.67±0.38	71.42±0.26	86.30±0.01	87.06±0.12	68.09±0.02	80.29±0.03	76.34±0.01	82.26±0.05	79.95±0.08
GATE-KD	93.47±0.12	74.18±0.25	72.50±0.47	85.45±0.08	87.24±0.12	66.57±0.08	81.85±0.06	76.40±0.07	82.52±0.08	80.02±0.03
(b) BGE-M3 → MiniLMv2-H768 (66M)										
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	79.80
Student base	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	56.36
RKD	91.68±0.06	74.57±0.40	62.45±0.75	85.81±0.08	79.81±0.17	63.57±0.13	76.91±0.05	69.54±0.11	75.56±0.09	75.54±0.10
Stella	88.72±0.41	73.43±0.23	61.63±0.29	86.39±0.08	74.13±0.31	66.98±0.27	73.32±0.07	68.30±0.12	72.66±0.14	73.95±0.04
CDM	87.64±0.23	70.82±0.14	56.37±0.57	85.67±0.08	67.69±0.23	66.89±0.20	70.15±0.48	66.68±0.26	69.49±0.15	71.27±0.10
DSKD	86.69±0.38	69.76±0.22	56.98±0.39	84.92±0.07	66.75±0.17	67.23±0.10	68.26±0.45	66.28±0.02	68.31±0.35	70.58±0.10
EMO	87.17±0.59	71.07±0.22	59.32±0.02	84.25±0.41	67.89±0.92	66.09±0.18	69.04±0.38	62.26±1.77	66.47±0.40	70.40±0.35
TALAS	92.44±0.10	75.58±0.15	65.13±0.21	87.09±0.09	85.39±0.14	64.39±0.12	78.14±0.06	72.56±0.13	81.03±0.05	77.97±0.03
GATE-KD	92.56±0.04	76.30±0.31	65.62±0.29	86.58±0.05	87.20±0.06	63.43±0.02	79.23±0.03	72.69±0.05	80.60±0.13	78.24±0.05
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)										
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	79.14
Student base	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	52.62
RKD	89.74±0.04	72.34±0.22	57.99±0.52	84.13±0.12	75.20±0.14	67.49±0.06	74.48±0.03	68.12±0.15	72.55±0.14	73.56±0.09
Stella	85.74±0.22	70.25±0.07	53.48±0.28	84.43±0.04	71.23±0.05	66.51±0.14	70.96±0.28	63.02±0.04	67.42±0.24	70.34±0.02
CDM	85.18±0.08	69.14±0.07	52.30±0.26	84.02±0.04	68.24±0.23	66.57±0.29	69.15±0.13	63.05±0.06	66.55±0.10	69.36±0.08
DSKD	85.04±0.14	69.21±0.13	52.85±0.36	84.34±0.10	67.33±0.24	65.98±0.12	69.01±0.13	63.46±0.21	66.42±0.09	69.29±0.09
EMO	85.08±0.17	68.11±0.10	53.96±0.60	84.38±0.16	65.48±0.16	65.23±0.17	68.61±0.07	60.00±0.82	65.38±0.38	68.47±0.15
TALAS	89.79±0.03	73.87±0.17	61.75±0.61	85.99±0.03	82.07±0.09	65.79±0.04	75.67±0.05	73.91±0.03	78.31±0.04	76.35±0.08
GATE-KD	91.68±0.09	74.43±0.21	66.88±0.36	85.00±0.06	83.46±0.02	67.44±0.04	77.94±0.06	72.17±0.02	78.36±0.07	77.48±0.04

Table 11 shows that GATE-KD attains the highest mean Avg. in all three teacher–student configurations. It exceeds TALAS by 0.07, 0.27, and 1.13 points in configurations (a), (b), and (c), respectively. The margin is narrow for the two larger students and remains largest for the compact student.

E.2 ZERO-SHOT RETRIEVAL

We evaluate the models distilled in the main comparison on ArguAna, SciFact, SciDocs, NFCorpus, and FiQA-2018 using exhaustive cosine ranking. Evaluation uses no instruction prefix or task-specific retrieval training and reports nDCG@10. Retrieval scores are not included in the sentence-level averages of Table 1. Teacher and student-base scores are deterministic, while distilled-student scores are mean $\pm$ sample standard deviation over three seeds.

Table 12: Zero-shot retrieval on five BEIR benchmarks (nDCG@10). Distilled-student scores are mean $\pm$ sample standard deviation over three seeds; best and second-best results are bold and underlined.

Method	ArguAna	SciFact	SciDocs	NFCorpus	FiQA-2018	Avg.
<i>(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)</i>						
Teacher	74.66	73.75	25.69	36.23	50.15	52.10
Student base	12.65	0.89	0.38	1.33	0.05	3.06
RKD	39.68 $\pm$ 0.25	9.74 $\pm$ 0.64	1.19 $\pm$ 0.04	4.30 $\pm$ 0.30	2.82 $\pm$ 0.09	11.55 $\pm$ 0.18
Stella	39.12 $\pm$ 0.17	14.29 $\pm$ 0.17	<u>3.79</u> $\pm$ 0.26	6.21 $\pm$ 0.14	5.57 $\pm$ 0.04	13.79 $\pm$ 0.03
CDM	33.62 $\pm$ 0.27	15.11 $\pm$ 0.20	3.28 $\pm$ 0.14	<u>6.26</u> $\pm$ 0.09	5.08 $\pm$ 0.22	12.67 $\pm$ 0.04
DSKD	33.70 $\pm$ 0.39	15.75 $\pm$ 0.49	3.35 $\pm$ 0.11	6.21 $\pm$ 0.17	5.34 $\pm$ 0.24	12.87 $\pm$ 0.27
EMO	21.90 $\pm$ 0.34	8.84 $\pm$ 0.28	1.87 $\pm$ 0.11	4.87 $\pm$ 0.16	2.04 $\pm$ 0.06	7.91 $\pm$ 0.07
TALAS	48.95 $\pm$ 0.17	19.94 $\pm$ 0.09	5.70 $\pm$ 0.12	7.94 $\pm$ 0.08	8.15 $\pm$ 0.02	18.14 $\pm$ 0.07
GATE-KD	<u>45.20</u> $\pm$ 0.32	<u>19.25</u> $\pm$ 0.66	2.81 $\pm$ 0.09	<u>6.79</u> $\pm$ 0.22	<u>6.56</u> $\pm$ 0.21	<u>16.12</u> $\pm$ 0.26
<i>(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)</i>						
Teacher	54.01	64.16	16.37	31.80	41.12	41.49
Student base	9.32	1.15	0.44	1.22	0.08	2.44
RKD	21.27 $\pm$ 0.23	7.84 $\pm$ 0.12	<u>1.96</u> $\pm$ 0.05	4.31 $\pm$ 0.14	2.26 $\pm$ 0.03	7.53 $\pm$ 0.09
Stella	18.46 $\pm$ 0.37	3.11 $\pm$ 0.27	1.25 $\pm$ 0.06	2.29 $\pm$ 0.15	1.66 $\pm$ 0.08	5.35 $\pm$ 0.16
CDM	15.83 $\pm$ 0.19	3.77 $\pm$ 0.38	0.88 $\pm$ 0.02	2.88 $\pm$ 0.16	1.41 $\pm$ 0.21	4.95 $\pm$ 0.15
DSKD	16.25 $\pm$ 0.15	4.05 $\pm$ 0.10	0.88 $\pm$ 0.03	2.96 $\pm$ 0.11	1.59 $\pm$ 0.07	5.15 $\pm$ 0.07
EMO	13.10 $\pm$ 0.27	2.83 $\pm$ 0.17	0.72 $\pm$ 0.04	2.25 $\pm$ 0.07	0.85 $\pm$ 0.09	3.95 $\pm$ 0.08
TALAS	30.14 $\pm$ 0.32	6.55 $\pm$ 0.34	1.90 $\pm$ 0.08	2.82 $\pm$ 0.18	<u>2.83</u> $\pm$ 0.09	8.85 $\pm$ 0.17
GATE-KD	<u>25.30</u> $\pm$ 0.02	11.22 $\pm$ 0.27	3.31 $\pm$ 0.02	4.93 $\pm$ 0.06	4.83 $\pm$ 0.01	9.92 $\pm$ 0.05
<i>(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)</i>						
Teacher	68.11	68.26	15.72	29.73	38.68	44.10
Student base	9.85	0.66	0.32	1.28	0.05	2.43
RKD	30.12 $\pm$ 0.79	<u>5.57</u> $\pm$ 0.76	<u>1.77</u> $\pm$ 0.12	<u>3.34</u> $\pm$ 0.07	<u>1.88</u> $\pm$ 0.06	8.54 $\pm$ 0.33
Stella	24.49 $\pm$ 0.45	2.65 $\pm$ 0.09	1.09 $\pm$ 0.07	2.62 $\pm$ 0.07	1.01 $\pm$ 0.11	6.37 $\pm$ 0.09
CDM	21.09 $\pm$ 0.30	1.66 $\pm$ 0.17	1.13 $\pm$ 0.03	2.57 $\pm$ 0.05	0.89 $\pm$ 0.02	5.47 $\pm$ 0.06
DSKD	21.56 $\pm$ 0.38	1.77 $\pm$ 0.20	1.18 $\pm$ 0.06	2.63 $\pm$ 0.03	0.89 $\pm$ 0.07	5.61 $\pm$ 0.13
EMO	17.18 $\pm$ 0.14	1.41 $\pm$ 0.09	0.92 $\pm$ 0.02	1.96 $\pm$ 0.06	0.70 $\pm$ 0.03	4.44 $\pm$ 0.05
TALAS	<u>35.21</u> $\pm$ 0.52	4.17 $\pm$ 0.09	1.44 $\pm$ 0.04	3.12 $\pm$ 0.10	1.87 $\pm$ 0.07	<u>9.16</u> $\pm$ 0.14
GATE-KD	36.18 $\pm$ 0.09	10.26 $\pm$ 0.21	3.44 $\pm$ 0.06	4.30 $\pm$ 0.06	3.57 $\pm$ 0.03	11.55 $\pm$ 0.06

GATE-KD achieves the highest retrieval average in configurations (b) and (c). It exceeds TALAS by 1.07 and 2.39 points, respectively, and leads on nine of the ten individual retrieval comparisons across those two configurations. In configuration (a), it ranks second to TALAS, with 16.12 versus 18.14 Avg. All distilled students remain substantially below their teachers, indicating that the sentence-level distillation objectives do not fully preserve the teachers’ zero-shot retrieval capability.

F REPRESENTATION DIAGNOSTICS

We examine whether final-embedding supervision changes the internal representations of the student even though GATE-KD applies no intermediate-layer objective. The probe contains 512 unique sentences sampled from the validation splits after excluding sentences found in the distillation corpus. For every teacher and student Transformer layer, we compute centered linear CKA between their pooled representations (Kornblith et al., 2019). Results for the trained students are averaged over seeds 42, 43, and 44. We omit the embedding-output layer because special-token pooling makes this representation constant across examples, for which centered CKA is undefined.

Figure 2 shows that the pretrained students already share substantial representation structure with their teachers. Raw CKA alone therefore combines correspondence inherited from pretraining with changes induced by distillation.

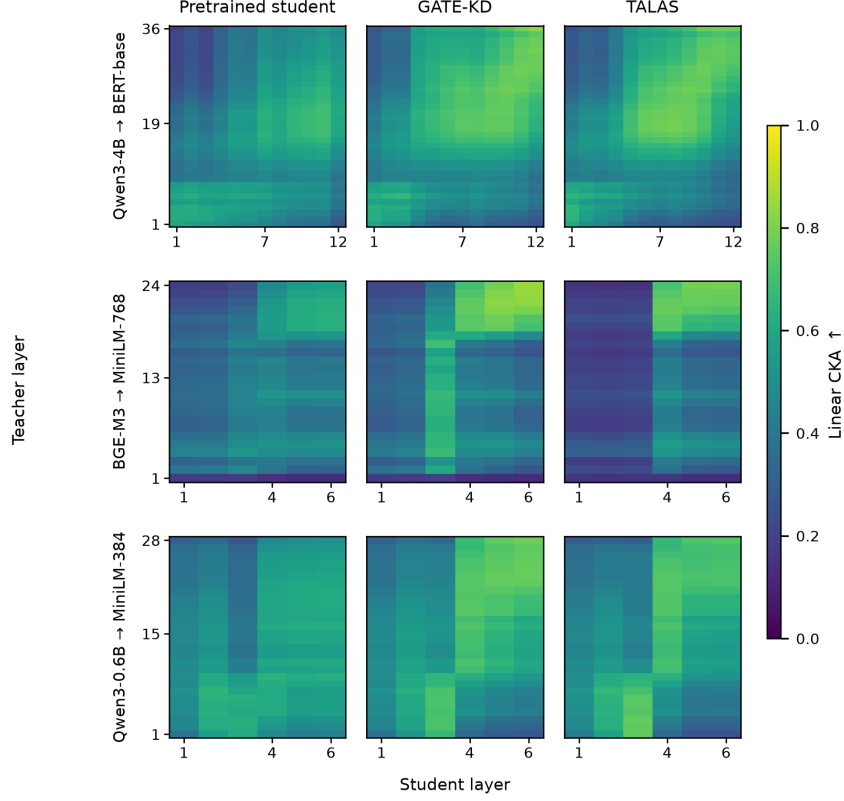


Figure 2: Layerwise linear CKA for the pretrained student, GATE-KD, and TALAS across three configurations. Trained results are averaged over three seeds.

To contextualize representation similarity relative to initialization, Figure 3 subtracts the pretrained-student CKA matrix from the mean matrix after distillation. GATE-KD produces higher final-to-final CKA than TALAS in all three configurations. The raw values are 0.8266, 0.8447, and 0.7845 for GATE-KD, compared with 0.7741, 0.7901, and 0.7333 for TALAS in configurations (a), (b), and (c), respectively. The corresponding changes from initialization are 0.2992, 0.2773, and 0.2781 for GATE-KD, compared with 0.2468, 0.2227, and 0.2269 for TALAS. GATE-KD consequently exhibits consistently higher final-to-final CKA across the three configurations, with differences ranging from 0.051 to 0.055.

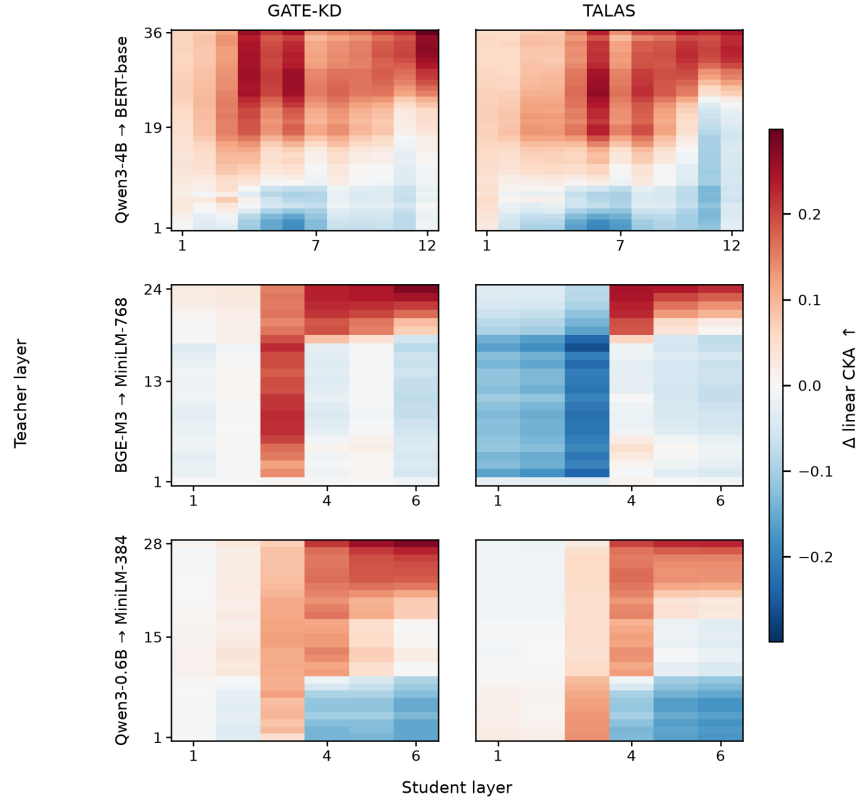


Figure 3: Change in layerwise linear CKA relative to the pretrained student. Red and blue denote increased and decreased similarity, respectively.

The changes are distributed across multiple later student layers rather than being confined to a single teacher–student layer pair. Thus final-embedding supervision co-occurs with measurable changes in later student representations despite the absence of explicit intermediate-layer matching. Some layer pairs decrease in CKA, so the result does not imply uniformly stronger alignment at every depth. CKA is descriptive: it does not establish that the observed changes cause downstream improvements or that endpoint supervision mechanistically substitutes for intermediate-layer distillation. It is also invariant to orthogonal transformations and isotropic scaling, so higher CKA measures representational similarity rather than transferred task information. The figures report seed means only; we have not measured seed-level variance of the CKA matrices, included a fixed-PCA control without a fitted gauge, or tested whether per-layer CKA changes correlate with downstream differences. We therefore use these probes to document that endpoint supervision co-occurs with changes in later layers, and for nothing stronger.